

Tianhao Wu

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Interests	Formal Methods, Control, Machine Learning, Safe Autonomy	
Education	University of Southern California	
	M.S. in Computer Science	Aug 2025 – Present
	B.S. in Computer Science and Computer Engineering	Jan 2021 – Dec 2024
	B.S. in Applied and Computational Mathematics GPA: 3.95/4.0	
Publications	An Algebraic Framework for Quantitative Semantics of Spatio-Temporal Logic with Graph Operators Sheryl Paul*, Vidisha Kudalkar*, Anand Balakrishnan*, Tianhao Wu , Lars Lindemann, Jyotirmoy V. Deshmukh <i>QEST + FORMATS 2026</i> [pdf]	
Preprints	Scenario MPC with STL Specifications and Pareto-Based Feasibility Repair Tianhao Wu , Yiwei Lyu <i>Under review at IEEE ICRA 2027</i> [pdf] [project]	
	Enhancing the Performance of DeepReach on High-Dimensional Systems through Optimizing Activation Functions Qian Wang*, Tianhao Wu * <i>arXiv 2023</i> [pdf]	
Research Experience	Research Assistant, University of Southern California	Apr 2026 – Present
	Advisor: Jyotirmoy V. Deshmukh Implemented algebraic quantitative semantics for Spatio-Temporal Logic with Graph Operators (STL-GO). Evaluated four semantics on 2D and 3D multi-agent environments. Investigating STL-GO specifications for credit assignment in multi-agent reinforcement learning (MARL).	
	Research Assistant, Texas A&M University	Oct 2025 – Present
	Advisor: Yiwei Lyu Developed a model predictive control (MPC) framework with Signal Temporal Logic (STL) specifications under stochastic and uncontrollable agents. Formulated feasibility repair as a Pareto optimization problem and derived a probabilistic certificate for STL violation rate. Evaluated the framework on autonomous-driving scenarios in CARLA with demonstrated safe behaviors.	

	Research Intern, Carnegie Mellon University Jan 2025 – Apr 2025 Advisor: John M. Dolan Designed a modular safe-control framework for autonomous driving. Applied Linear Temporal Logic (LTL) specifications for dynamic activation of control barrier function (CBF) constraints. Implemented preliminary safety modules in CARLA.
	Research Assistant, University of Southern California Aug 2022 – May 2023 Advisor: Somil Bansal Investigated the performance of DeepReach on high-dimensional Hamilton–Jacobi (HJ) reachability problems with alternative activations. Reduced violation rates by 5.1% on a 9D multi-vehicle collision-avoidance problem.
Honors & Awards	Ming Hsieh Institute Undergraduate Scholar 2023 – 2024 One of the five scholars selected for research potential USC Academic Achievement Award Fall 2023 Viterbi CURVE Research Fellowship 2022 – 2023 Lenore B. Kreiger Scholarship for Mathematics 2022 – 2023
Relevant Coursework	PhD-level: Robot Learning; Automata Theory, Logic and Formal Verification Graduate-level: Applied Probability; Autonomous Cyber-Physical Systems; Learning and Control for Safety-Critical Robotic Systems
Teaching Experience	Teaching Assistant, University of Southern California Discrete Methods in Computer Science (CS170) Fall 2026 Fundamentals of Computation (CS102), Head TA Spring 2022 – Spring 2024 Introduction to Embedded Systems (EE109) Fall 2023 Introduction to Artificial Intelligence (CS360) Spring 2023, Fall 2026 CS@SC Summer Coding Camp Summer 2022
Service	Reviewer , Learning and Formal Methods for Robotics, IROS Workshop (2026)
Skills	Programming: Python, C++, MATLAB Tools: Gurobi, CVXPY, PyTorch, CARLA, Linux, Git, LaTeX